Class 09: Structural Bioinformatics (pt. 1)

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The PDB Statistics

The main database for structural biology is called the PDB. Let's have a look at what it contains:

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
data <- read.csv(file="Data Export Summary.csv")
head(data)
```

```
           Molecular.Type  X.ray     EM    NMR Integrative Multiple.methods
1          Protein (only) 176,204 20,299 12,708         342              218
2 Protein/Oligosaccharide  10,279  3,385     34           8               11
3              Protein/NA   9,007  5,897    287          24                7
4     Nucleic acid (only)   3,066    200  1,553           2               15
5                   Other     173     13     33           3                0
6  Oligosaccharide (only)      11      0      6           0                1
  Neutron Other   Total
1      83    32 209,886
2       1     0  13,718
3       0     0  15,222
4       3     1   4,840
```

```
5        0      0      222
6        0      4       22
```

```r
data$Total
```

```
[1] "209,886" "13,718"  "15,222"  "4,840"   "222"     "22"
```

```r
data$Neutron
```

```
[1] 83  1  0  3  0  0
```

Some characters are not numeric

```r
library(tidyverse)
```

```r
pdb <- read_csv("Data Export Summary.csv")
pdb
```

```
# A tibble: 6 x 9
  `Molecular Type`     `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
  <chr>                  <dbl> <dbl> <dbl>       <dbl>              <dbl>   <dbl>
1 Protein (only)        176204 20299 12708         342                218      83
2 Protein/Oligosacch~    10279  3385    34           8                 11       1
3 Protein/NA              9007  5897   287          24                  7       0
4 Nucleic acid (only)     3066   200  1553           2                 15       3
5 Other                    173    13    33           3                  0       0
6 Oligosaccharide (o~        11     0     6           0                  1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

```r
pro.xray <- sum(pdb$`X-ray`)/sum(pdb$Total)*100
pro.em <- sum(pdb$EM)/sum(pdb$Total)*100
round(pro.xray,2)
```

```
[1] 81.48
```

```r
round(pro.em,2)
```

```
[1] 12.22
```

81.48% of structures in PDB are solved by x-ray, and 12.22% of structures in PDB are solved by Electron Microscopy.

> Q2: What proportion of structures in the PDB are protein?

```
pro.protein <- sum(pdb[1:3,"Total"])/sum(pdb$Total)*100
round(pro.protein,2)
```

```
[1] 97.92
```

97.92% of structures in the PDB are protein.

Exploring PDB Statistics

> Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

4,865 structures are currently in PDB.

Package for structural bioinformatics

```
library(bio3d)

hiv <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

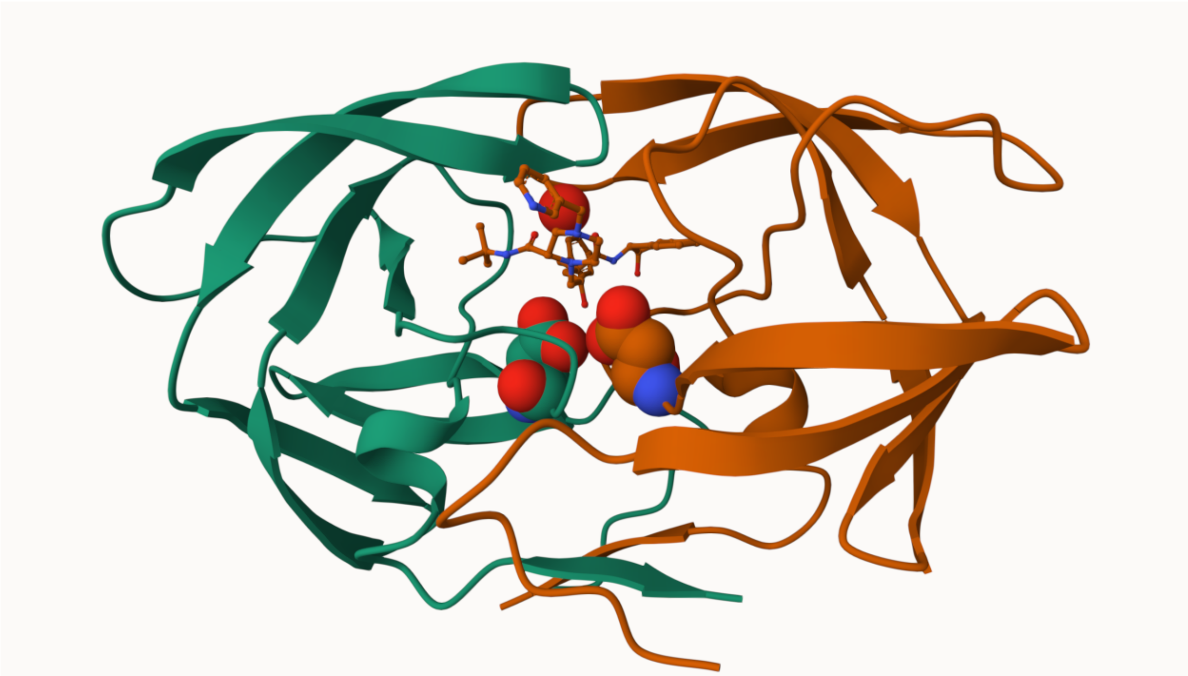
```
hiv
```

```

 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

Let's first use the Mol* viewer to explore this structure.

Figure 1: First view of HIV-pr



PDB objects in R

```
head(hiv$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

Extract the sequence

```r
pdbseq(hiv)
```

```
  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20 
"P" "Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" 
 21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40 
"E" "A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" 
 41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60 
"R" "W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" 
 61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80 
"Q" "I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" 
 81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99   1 
"P" "V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" "P" 
  2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20  21 
"Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" "E" 
 22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40  41 
"A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" "R" 
 42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60  61 
"W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" "Q" 
 62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  81 
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P" 
 82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99 
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" 
```

```r
chainA_seq <- pdbseq(trim.pdb(hiv, chain="A"))
```

I can interactively view these PDB objects with the **bio3dviewer** package.

```r
library(bio3dview)
# view.pdb(hiv)
```

```r
# sel <- atom.select(hiv,resno=25)

# view.pdb(hiv, highlight = sel,
         # highlight.style = "spacefill",
         # colorScheme = "chain",
         # col=c("blue","red"),
         # backgroundColor = "grey")
```

Predicting Protein Flexibility

We can run a bioinformatics calculation to predict protein dynamics- i.e. functional motions.

We will use the `nma()` function

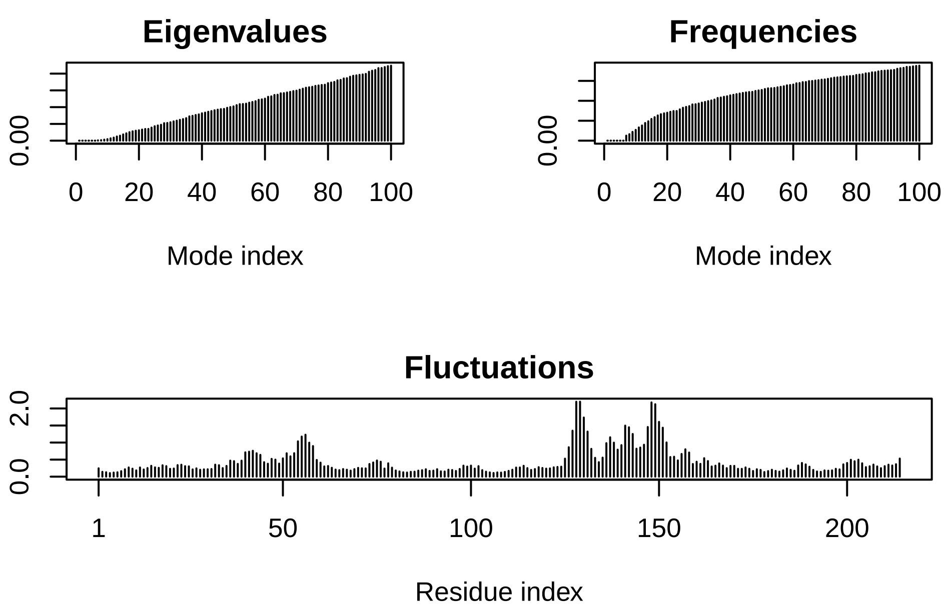
```
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.03 seconds.
 Diagonalizing Hessian...   Done in 0.53 seconds.
```

```
plot(m)
```



Generate a "trajectory" of predicted motion

```r
mktrj(m, file="ADK_nma.pdb")
```